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GLOBAL SYSTEM FOR
MOBILE COMMUNICATIONS

# Digital cellular telecommunications system; Unstructured Supplementary Service Data (USSD) - Stage 2 (GSM 03.90)

**ETSI**

European Telecommunications Standards Institute

**ETSI Secretariat**

**Postal address:** F-06921 Sophia Antipolis CEDEX - FRANCE
**Office address:** 650 Route des Lucioles - Sophia Antipolis - Valbonne - FRANCE
**X.400:** c=fr, a=atlas, p=etsi, s=secretariat - **Internet:** secretariat@etsi.fr

Tel.: +33 4 92 94 42 00 - Fax: +33 4 93 65 47 16

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## Foreword

This Global System for Mobile communications Technical Specification (GTS) has been produced by the Special Mobile Group (SMG) Technical Committee (TC) of the European Telecommunications Standards Institute (ETSI).

This GTS defines the stage 2 description of Unstructured Supplementary Service Data (USSD) within the digital cellular telecommunications system.

The contents of this GTS are subject to continuing work within TC-SMG and may change following formal TC-SMG approval. Should TC-SMG modify the contents of this GTS it will then be republished by ETSI with an identifying change of release date and an increase in version number as follows:

Version 5.x.y

where:

- y the third digit is incremented when editorial only changes have been incorporated in the specification;

- x the second digit is incremented for all other types of changes, i.e. technical enhancements, corrections, updates, etc.

The specification from which this GTS has been derived was originally based on CEPT documentation, hence the presentation of this GTS may not be entirely in accordance with the ETSI/PNE rules.

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# 1 Scope

This Global System for Mobile communications Technical Specification (GTS) gives the stage 2 description of Unstructured Supplementary Service Data (USSD).

The unstructured supplementary service data (USSD) mechanism allows the Mobile Station (MS) user and a PLMN operator defined application to communicate in a way which is transparent to the MS and to intermediate network entities. The mechanism allows development of PLMN specific supplementary services. The following diagram shows how handling of USSD is carried out, independently of the applications.

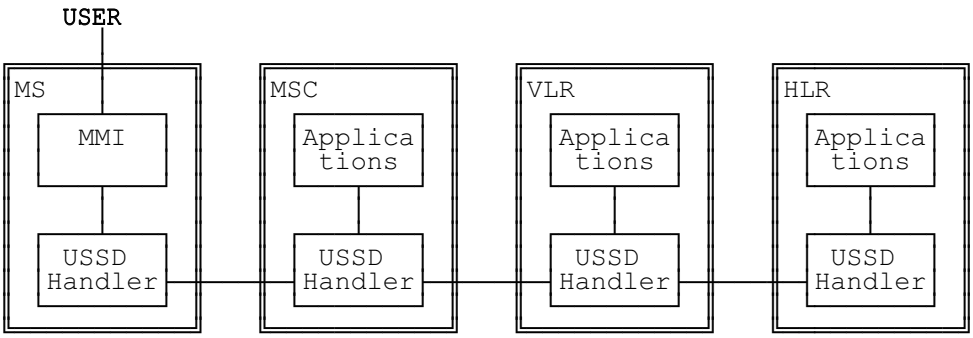



**Figure 1.1: Handling of USSD**

This specification defines the requirements for handling USSD at the MS and network entities. It does not include specification of particular applications, nor does it specify how a particular application is selected. Where more than one application exists at a network entity, routing of messages to the correct application is carried out by the USSD handler. The MMI for USSD is specified in GSM 02.30 and GSM 02.90. The alphabet indicator and the data coding scheme are defined in GSM 03.38.

USSD may be initiated by the MS user, or by the network in the following ways:

- Network initiated USSD (clause 1);

- Mobile initiated USSD (clause 2).

# 2 Normative references

This GTS incorporates by dated and undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this GTS only when incorporated in it by amendment or revision. For undated references, the latest edition of the publication referred to applies.

[1] GSM 01.04 (ETR 350): "Digital cellular telecommunications system (Phase 2+); Abbreviations and acronyms".

[2] GSM 02.30 (ETS 300 907): "Digital cellular telecommunications system (Phase 2+); Man-Machine Interface (MMI) of the Mobile Station (MS)".

[3] GSM 02.90: "Digital cellular telecommunications system; Unstructured Supplementary Service Data (USSD) - Stage 1".

[4] GSM 03.38 (ETS 300 900): "Digital cellular telecommunications system (Phase 2+); Alphabets and language-specific information".

# 3 Abbreviations

In addition to those below, abbreviations used in this GTS are listed in GSM 01.04 [1].

| | |
|---|---|
| AI | Application Initiated |
| MI | Mobile Initiated |
| USSD | Unstructured Supplementary Service Data |

# 4 Cross phase compatibility

The Phase 1 series of GSM specifications defined the signalling protocol which may be used, but they did not specify the operation of USSD as a service.

The main body of this specification assumes that the MS and all network entities comply with this phase of USSD. In order to minimize any possible problems between a Phase 1 implementation of USSD and this phase, subclauses 5.6 and 6.6 define the additional requirements for when one or more entity complies with the Phase 1 USSD specification for network initiated and mobile initiated USSD respectively.

# 5 Network initiated unstructured supplementary service

## 5.1 Handling of network initiated USSD

The network (MSC, VLR or HLR) can at any time send a USSD operation towards an MS. This operation may be either a request (asking the MS to provide information) or a notification (requiring no information in the response from the MS). No prior provision of USSD is required, although provision of services which make use of USSD may be required. All USSD requests, notifications and responses (except responses to notifications) contain the USSD string, an alphabet indicator and language indicator.

## 5.2 Functions and information flows

The following text describes the handling of network initiated USSD. Diagrammatic representations are as follows:

Figure 5.1 SDL for USSD invocation (HLR, VLR, MSC);
Figure 5.2 SDL for forwarding of USSD operations (VLR, MSC);
Figure 5.3 SDL for MS;
Figure 5.4 Information flow for successful single USSD request;
Figure 5.5 Information flow for successful single USSD notification;
Figure 5.6 Information flow for successful multiple USSD requests;
Figure 5.7 Information flow for failed USSD request.

### 5.2.1 Invoking unstructured SS operation from the HLR

When an application in the HLR is to send a USSD request or notification to an MS, it shall set up a transaction to the VLR where the subscriber is currently registered and send the operation to the VLR. It shall then await a response. The HLR is responsible for controlling the transaction, and shall therefore normally release the transaction when it receives a response from the VLR. The HLR may also release the transaction before receiving a response if necessary (e.g. if an application timer expires).

If an application in the HLR needs to send further operations to the same MS as part of the same application, it may continue to use the same transaction until all operations are completed (see figure 5.6). If a different transaction is to be used for a subsequent operation, the HLR shall release the first transaction before starting the next.

If the VLR releases the transaction at any time (e.g. due to user clearing), the HLR shall inform the application and terminate the USSD operation.

See subclause 5.2.4 for forwarding of an HLR invoked operation by the VLR and MSC.

### 5.2.2 Invoking unstructured SS operation from the VLR

When an application in the VLR is to send a USSD request or notification to an MS, it shall set up a transaction to the MSC where the subscriber is currently registered and send the operation to the MSC. It shall then await a response. The VLR is responsible for controlling the transaction, and shall therefore normally release the transaction when it receives a response from the MSC. The VLR may also release the transaction before receiving a response if necessary (e.g. if an application timer expires).

If an application in the VLR needs to send further operations to the same MS as part of the same application, it may continue to use the same transaction until all operations are completed. If a different transaction is to be used for a subsequent operation, the VLR shall release the first transaction before starting the next.

See subclause 5.2.4 for forwarding of a VLR invoked operation by the MSC.

If the MSC releases the transaction at any time (e.g. due to the user clearing), the VLR shall inform the application and terminate the USSD operation.

### 5.2.3 Invoking unstructured SS operation from the MSC

When an application in the MSC is to send a USSD request or notification to an MS, it shall set up a transaction to the MS where the subscriber is currently registered and send the operation to the MS. It shall then await a response. The MSC is responsible for controlling the transaction, and shall therefore normally release the transaction when it receives a response from the MS. The MSC may also release the transaction before receiving a response if necessary (e.g. if an application timer expires).

If an application in the MSC needs to send further operations to the same MS as part of the same application, it may continue to use the same transaction until all operations are completed. If a different transaction is to be used for a subsequent operation, the VLR shall release the first transaction before starting the next.

If the MS releases the transaction at any time (e.g. due to the user clearing), the MSC shall inform the application and terminate the USSD operation.

NOTE: MSC invoked USSD is only likely to be used for call related operations, where the application is controlling a call to or from the MS.

### 5.2.4 Forwarding USSD operations

The VLR may any time receive a USSD operation from the HLR. If the subscriber can be contacted, the VLR shall set up a transaction to the MSC and forward the operation unchanged. Any further information exchange between the HLR and MSC shall be transparent to the VLR. When one transaction is released, the VLR shall release the other.

The MSC may at any time receive an USSD operation from the VLR. If the subscriber can be contacted, the MSC shall set up a transaction to the MS and forward the operation unchanged. Any further information exchange between the VLR and MS shall be transparent to the MSC. When one transaction is released, the MSC shall release the other.

### 5.2.5 Handling of unstructured SS operation at the MS

The MS may at any time receive a USSD operation (request or notification) from the MSC.

If the MS receives a USSD transaction while another USSD transaction (network or mobile initiated) or a non-call related supplementary service transaction is in progress, the MS shall reject the new transaction.

If the MS receives a USSD operation when it is in a state where the MMI required is not possible (e.g. during dialling) it shall reject the operation.

If the MS does not support the alphabet indicated in the USSD operation, it shall inform the network.

If the MS is in a state where it can handle the operation, it shall process the operation as follows:

- For a USSD request, the MS shall display the text provided and await user input. If the user enters a response, the MS shall return the response to the MSC, maintaining the transaction. If the user requests release of the transaction, the MS shall release the transaction. See GSM 02.30 for details of the MMI.

- For a USSD notification, the MS shall display the text provided and send back a response.

After sending the response to a USSD operation, the MS shall wait for the network to release the transaction. If, while awaiting this release, the MS receives any further USSD operations, it shall process them in the normal way.

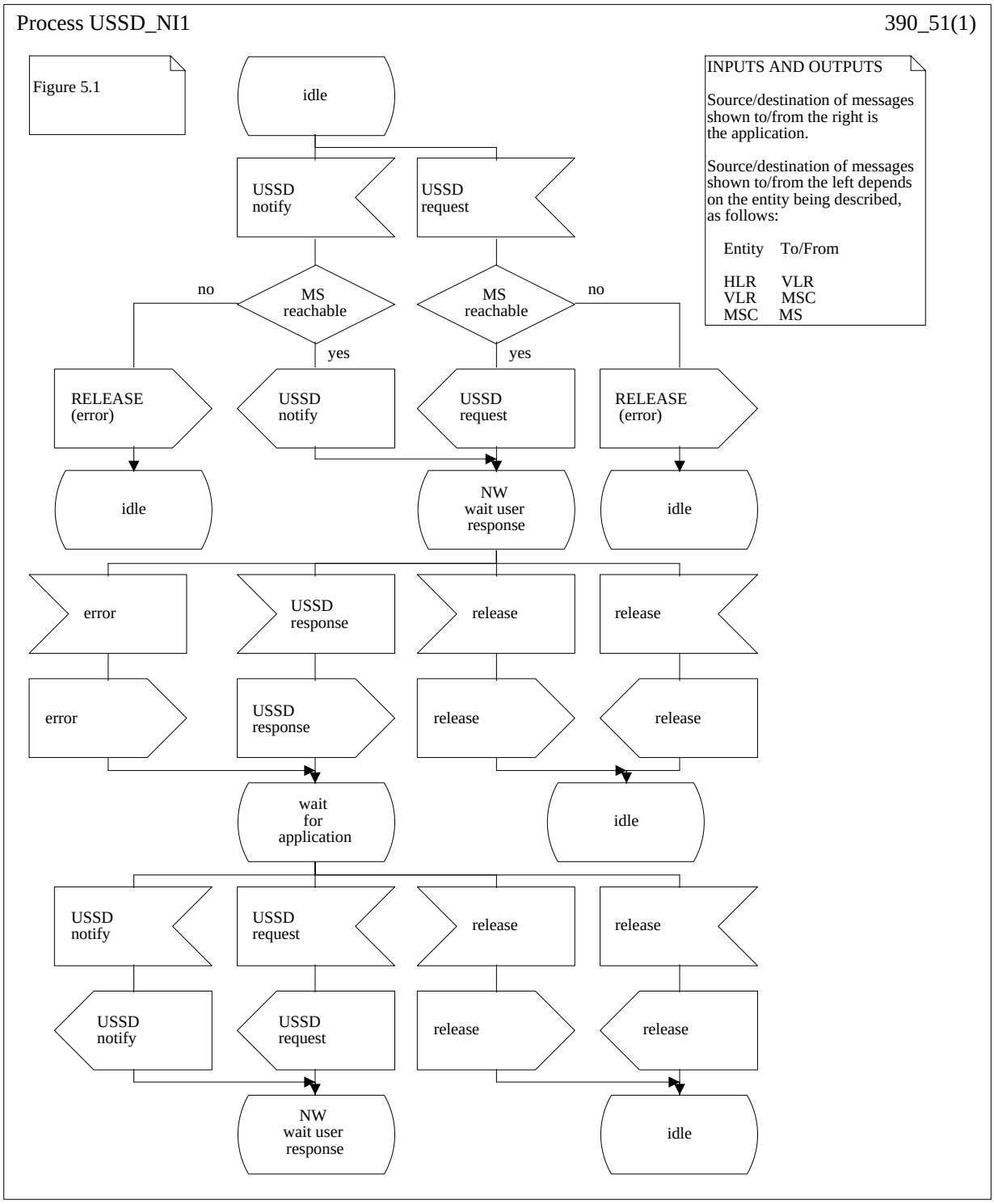



**Figure 5.1: Network initiated USSD invoked at HLR, VLR or MSC**

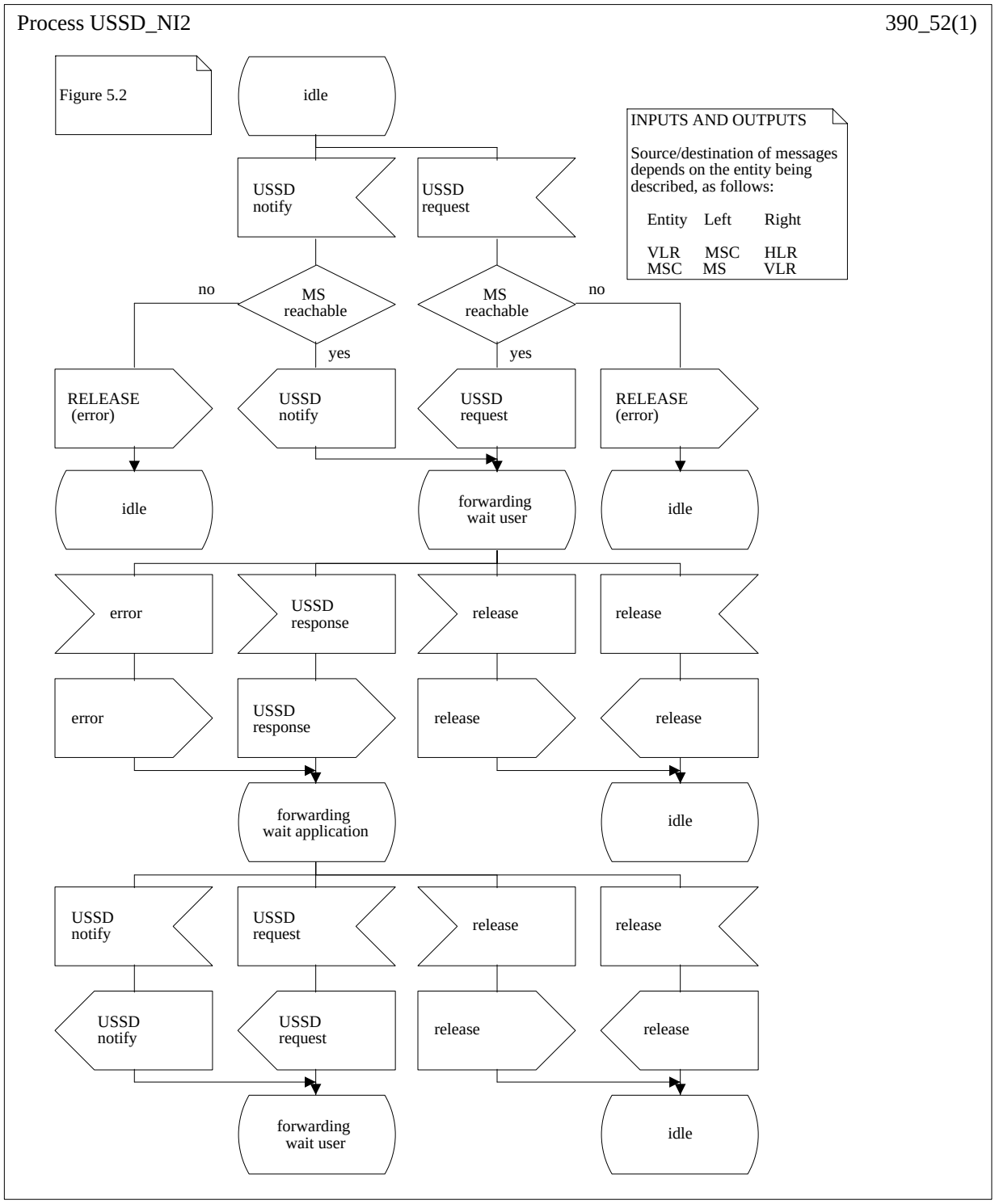



**Figure 5.2: Network initiated USSD forwarding at VLR or MSC**

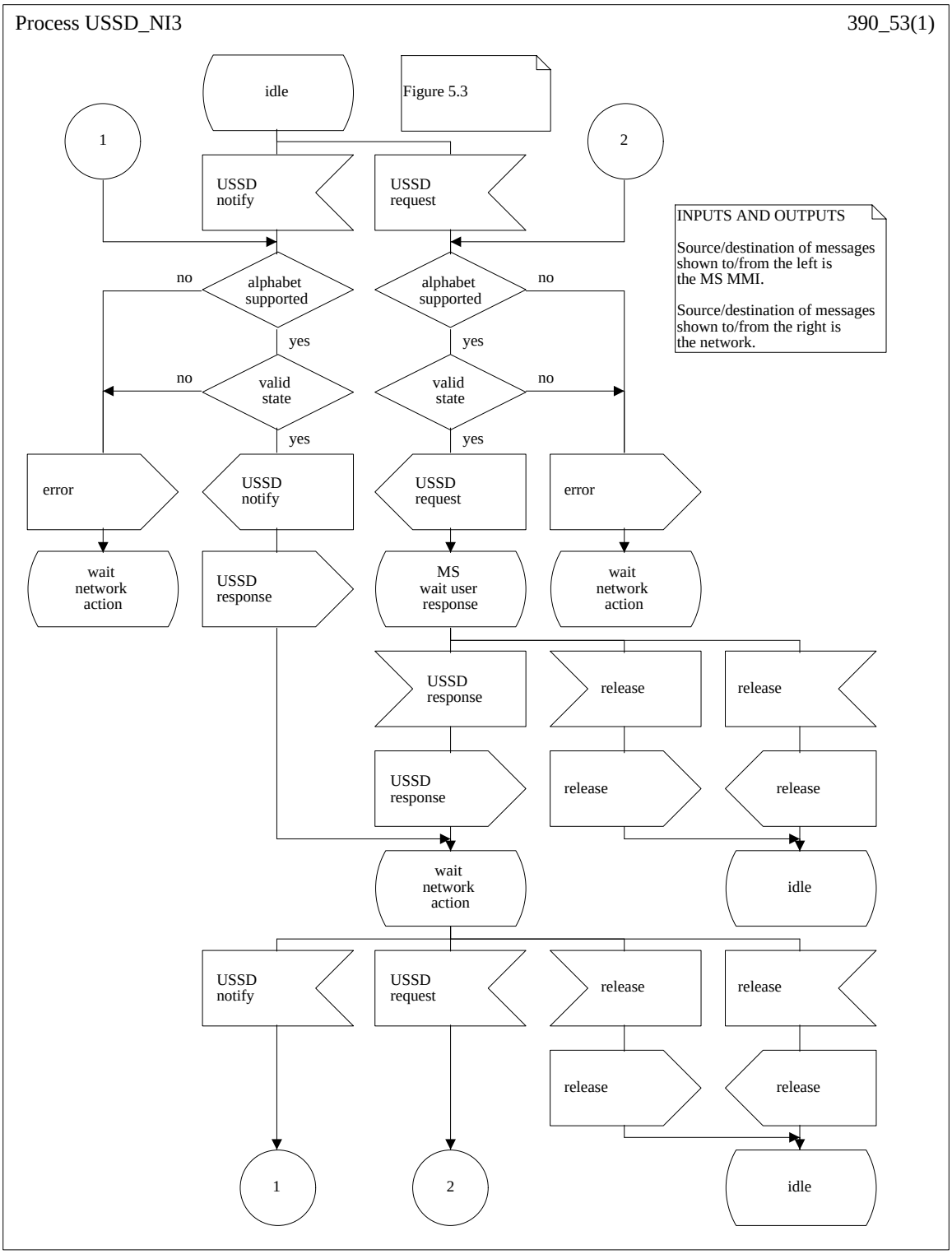



**Figure 5.3: Network initiated USSD at MS**

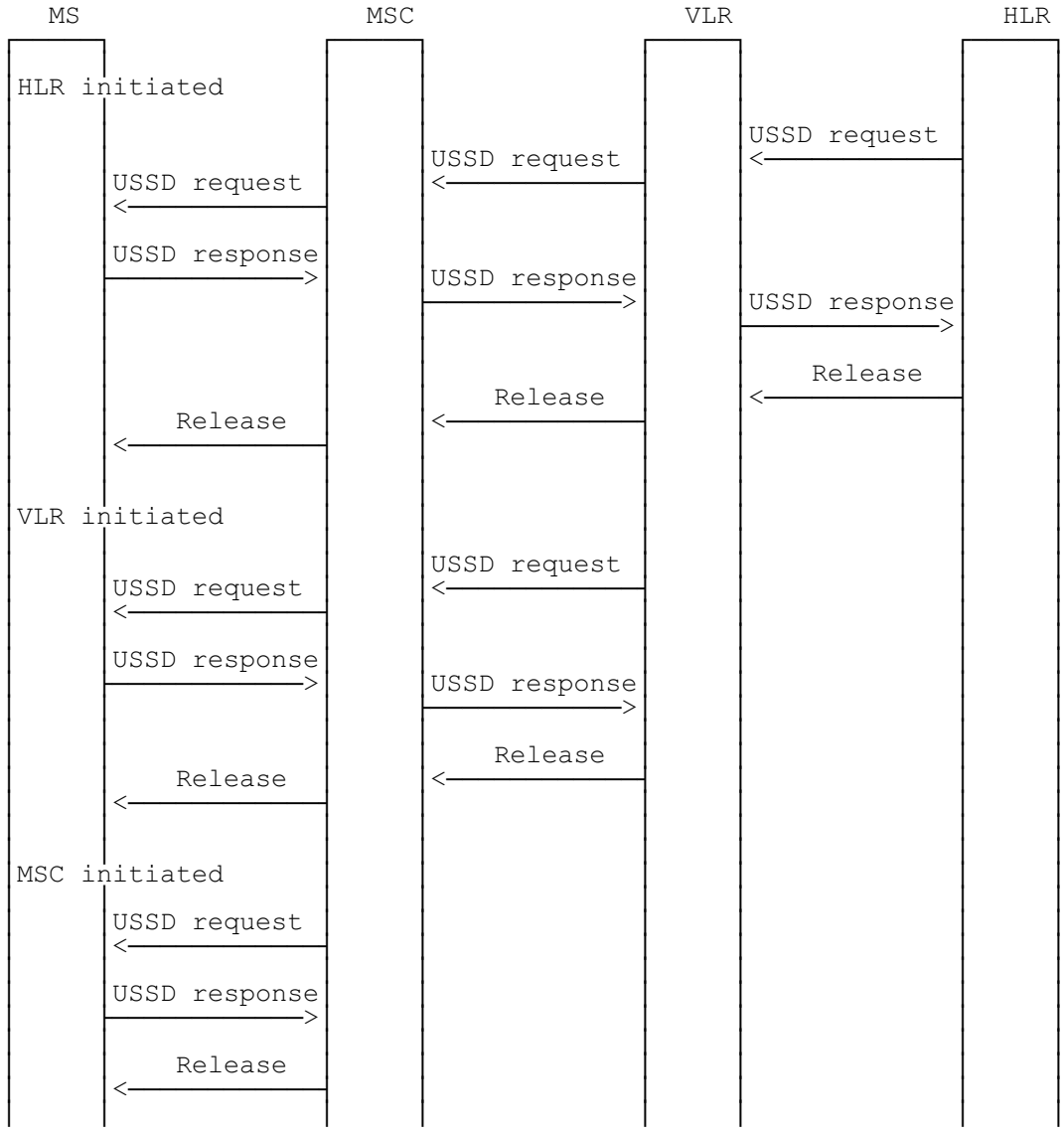



**Figure 5.4: Information flow for USSD request (Single Operation)**

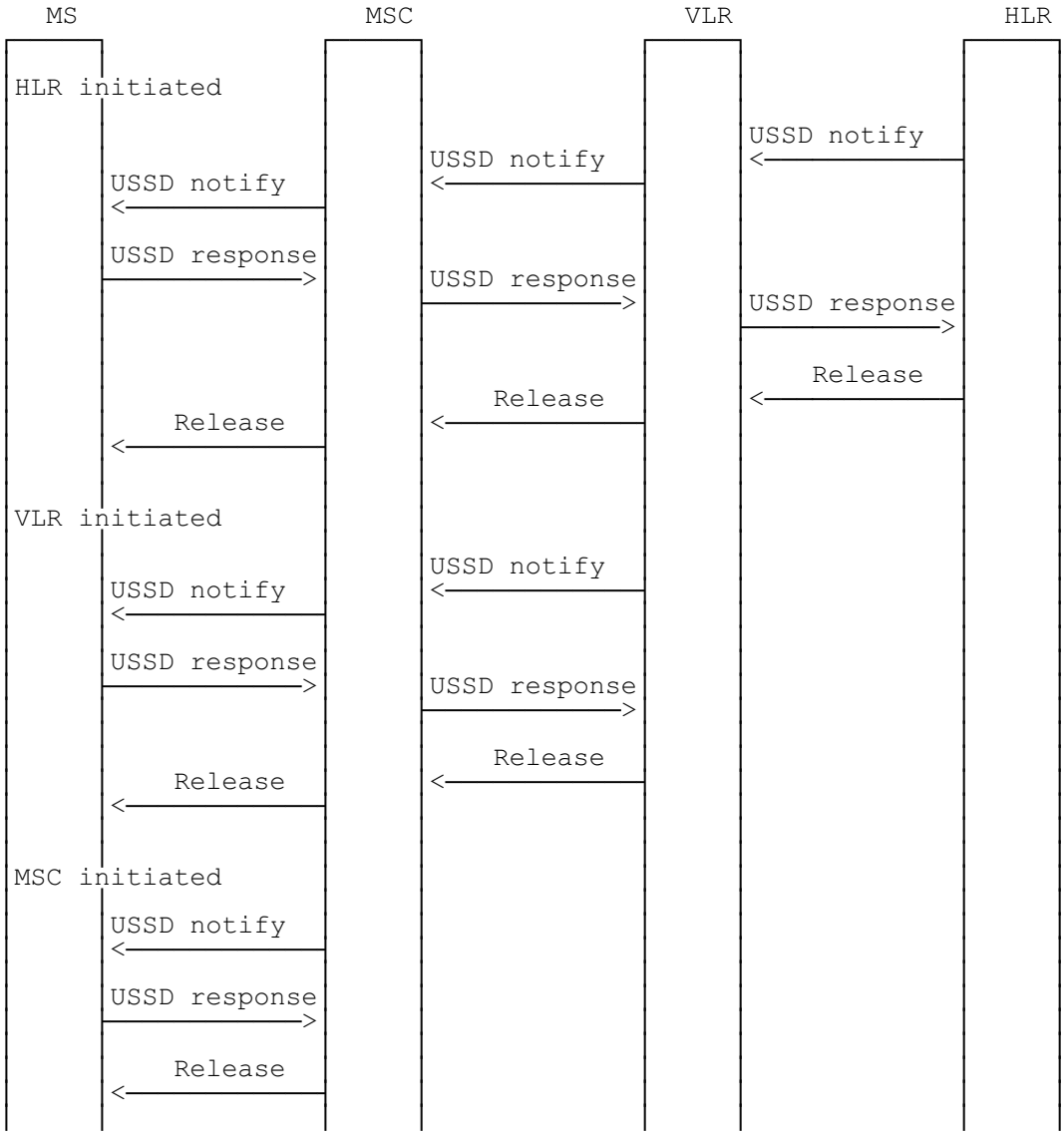



**Figure 5.5: Information flow for USSD notification (Single Operation)**

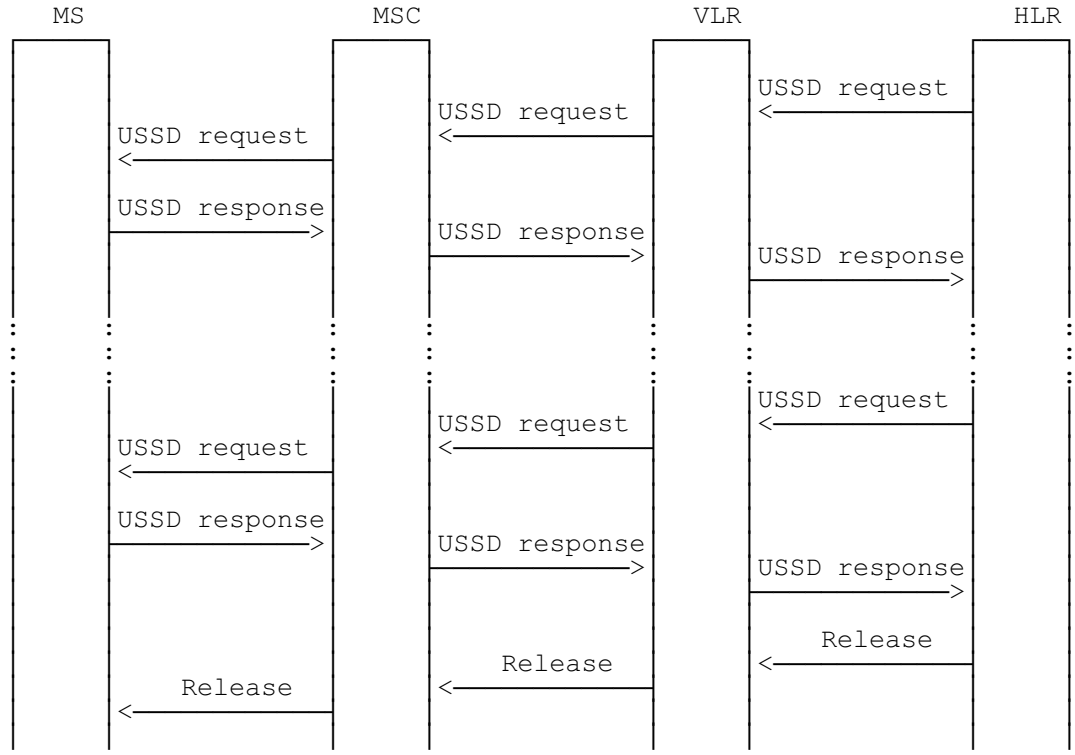



**Figure 5.6: Information flow for HLR initiated USSD request (Multiple Operation On Same Transaction)**

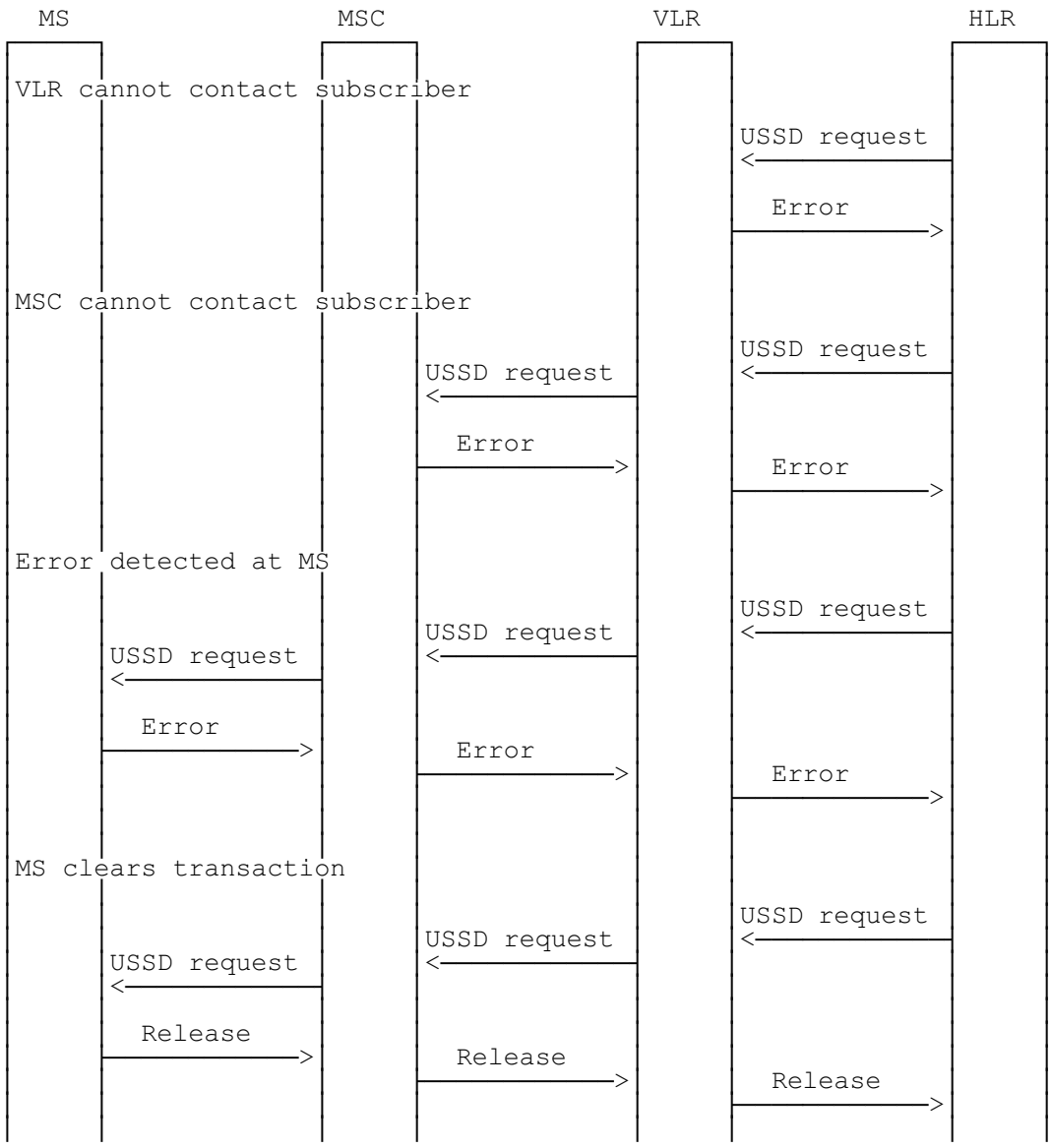



**Figure 5.7: Information flow for failed USSD request**

### 5.3 Information stored in the HLR

The HLR shall not store any information specific to the use of USSD, although information may be stored for services which are offered by USSD applications.

### 5.4 Information stored in the VLR

The VLR shall not store any information specific to the use of USSD, although information may be stored for services which are offered by USSD applications.

### 5.5 Handover

Handover will have no impact on the operation of this service.

### 5.6 Cross-phase compatibility

Network initiated USSD shall not be permitted if the MS or any network entity involved in the operation is of Phase 1. If, when setting up a transaction, a network entity discovers that the other end is of Phase 1, it shall reject the request and release the transaction being set up.

# 6 Mobile initiated unstructured supplementary service data

## 6.1 Handling of mobile initiated USSD

A MS can at any time initiate a USSD request to the network. No prior provision of the service is required, although provisioning of services which make use of USSD may be required. All USSD messages (requests and responses), contain the USSD string, an alphabet indicator and language indicator.

## 6.2 Functions and information flows

The following text describes the handling of mobile network initiated USSD. Diagrammatic representations are as follows:

Figure 6.1 SDL, request from user at MS;

Figure 6.2 SDL, request from MS at MSC;

Figure 6.3 SDL, request from application at MSC;

Figure 6.4 SDL, request from MSC at VLR;

Figure 6.5 SDL, request from application at VLR;

Figure 6.6 SDL, request from VLR at HLR;

Figure 6.7 Information flow, no further information required;

Figure 6.8 Information flow, further information required;

Figure 6.9 Information flow for failed USSD request.

### 6.2.1 Handling of USSD request at MS

When the user makes a request which the MS determines is to make use of USSD, the MS shall set up a transaction to the network, send the request to the MSC and await a response. When the MS receives the response, it shall display the information contained to the user.

While awaiting the response, the MS may receive a network initiated USSD request or notification on the same transaction. If this occurs, the MS shall process that operation (see section 1) and continue to await the response to the mobile initiated request.

If, when the MS determines that a user request is to make use of USSD, the MS is already involved in a USSD or a non-call related supplementary service transaction, the MS shall reject the request.

### 6.2.2 Handling of USSD request at MSC

When an MSC receives a USSD request containing an HPLMN service code, it shall set up a transaction to the VLR and forward the request unchanged. If this forwarding fails, an error shall be returned to the MS. The MSC shall be transparent to any further requests or responses (in either direction) for that transaction, passing them between the MS and VLR without taking any action. When one transaction is released (MS-MSC or MSC-VLR), the MSC shall release the other.

If an HPLMN service code is not included, the MSC shall process the request locally (see section 6.2.5).

If the MSC does not support the alphabet used in a USSD request, it shall set up a transaction to the VLR and forward the request unchanged, in the same way as when a HPLMN service code is received.

### 6.2.3 Handling of USSD request at VLR

When a VLR receives a USSD request containing an HPLMN service code and the user is not in the HPLMN, it shall set up a transaction to the HLR and forward the request unchanged. If this forwarding fails, an error shall be returned to the MS. The VLR shall be transparent to any further requests or responses (in either direction) for that transaction, passing them between the MSC and HLR without taking any action. When one transaction is released (MSC-VLR or VLR-HLR), the VLR shall release the other.

If an HPLMN service code is not included, or the user is in the HPLMN, the VLR shall process the request locally (see subclause 6.2.5).

If the VLR does not support the alphabet used in a USSD request, it shall set up a transaction to the HLR and forward the request unchanged, in the same way as when a HPLMN service code is received and the user is not in the HPLMN.

### 6.2.4 Handling of USSD request at HLR

An HLR shall always process a USSD request locally (see subclause 6.2.5).

If the HLR does not support the alphabet used in a USSD request, it shall inform the MS and release the transaction.

### 6.2.5 Processing the USSD request

When a network entity is to process a USSD request locally, the request shall be handled by an appropriate application. The location, nature and contents of USSD applications is, by definition, service provider and network operator dependent, but may include:

- setting up or releasing signalling and/or speech channels;

- passing the request to another network entity (unchanged or changed);

- passing a different USSD request to another network entity;

  and/or

- requesting further information from the MS (one or more times).

Upon completion of handling the request, the network entity shall respond to the request and release the transaction.

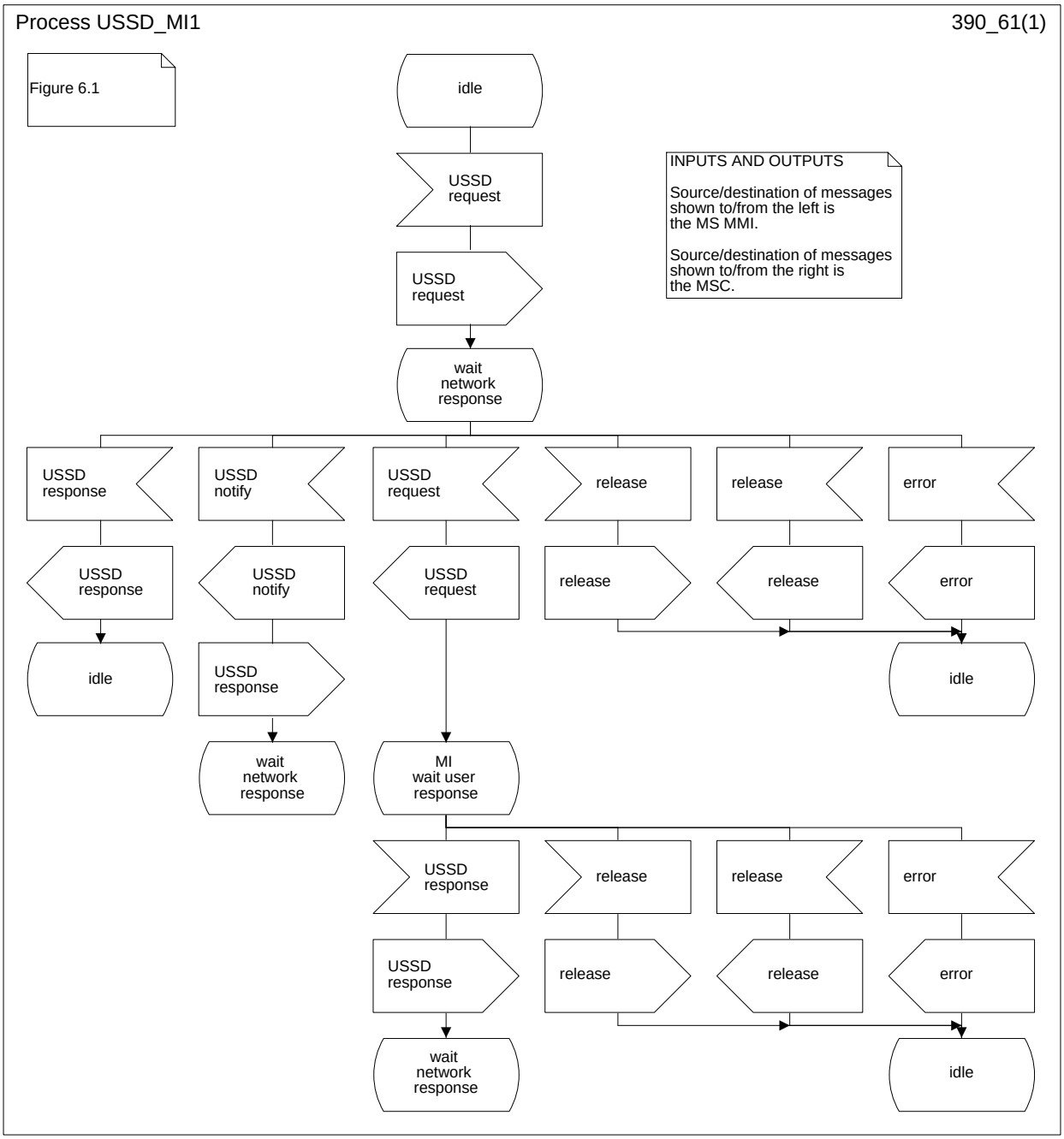



**Figure 6.1: Mobile initiated USSD at MS**

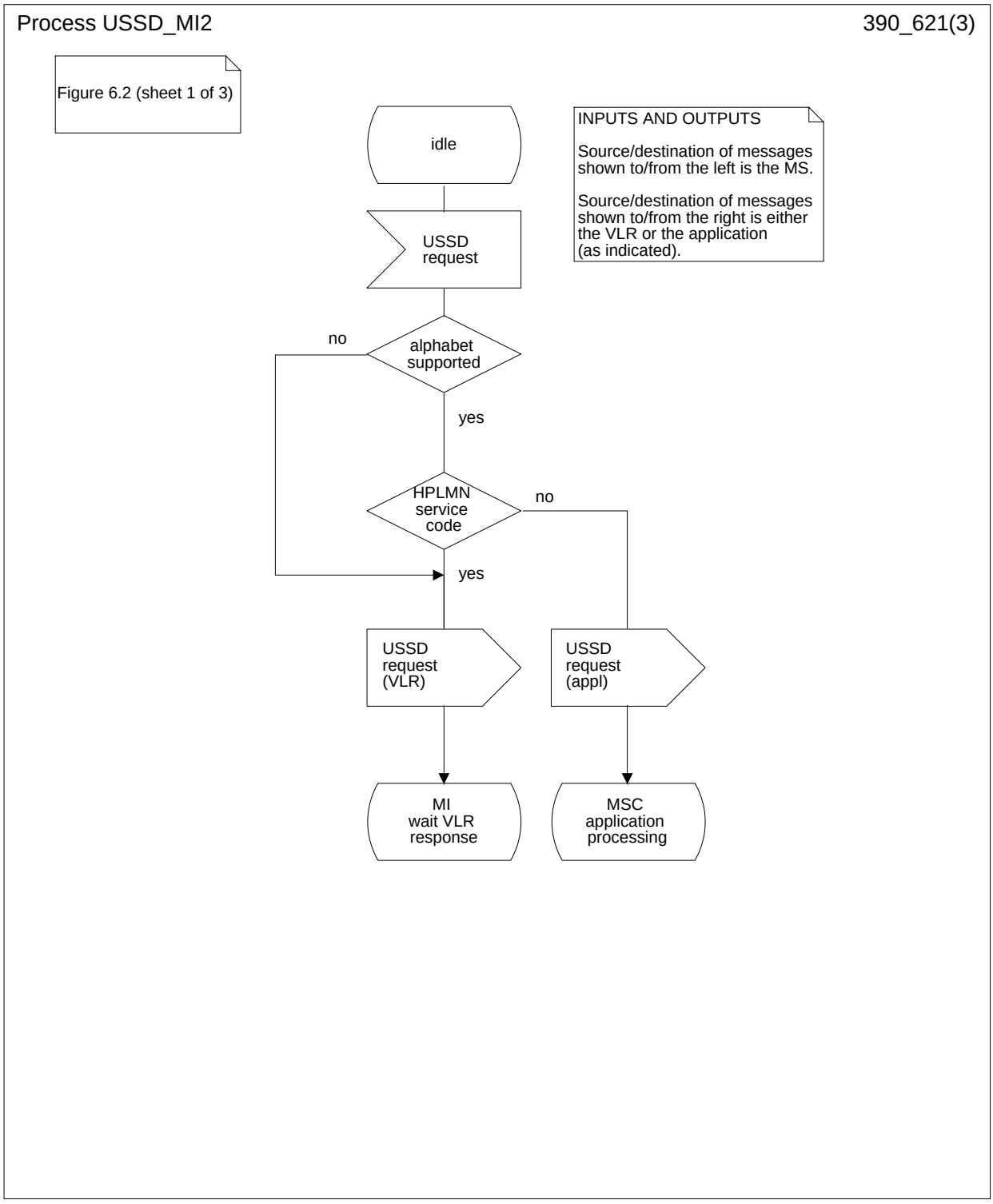



**Figure 6.2 (sheet 1 of 3): Mobile initiated USSD at MSC**

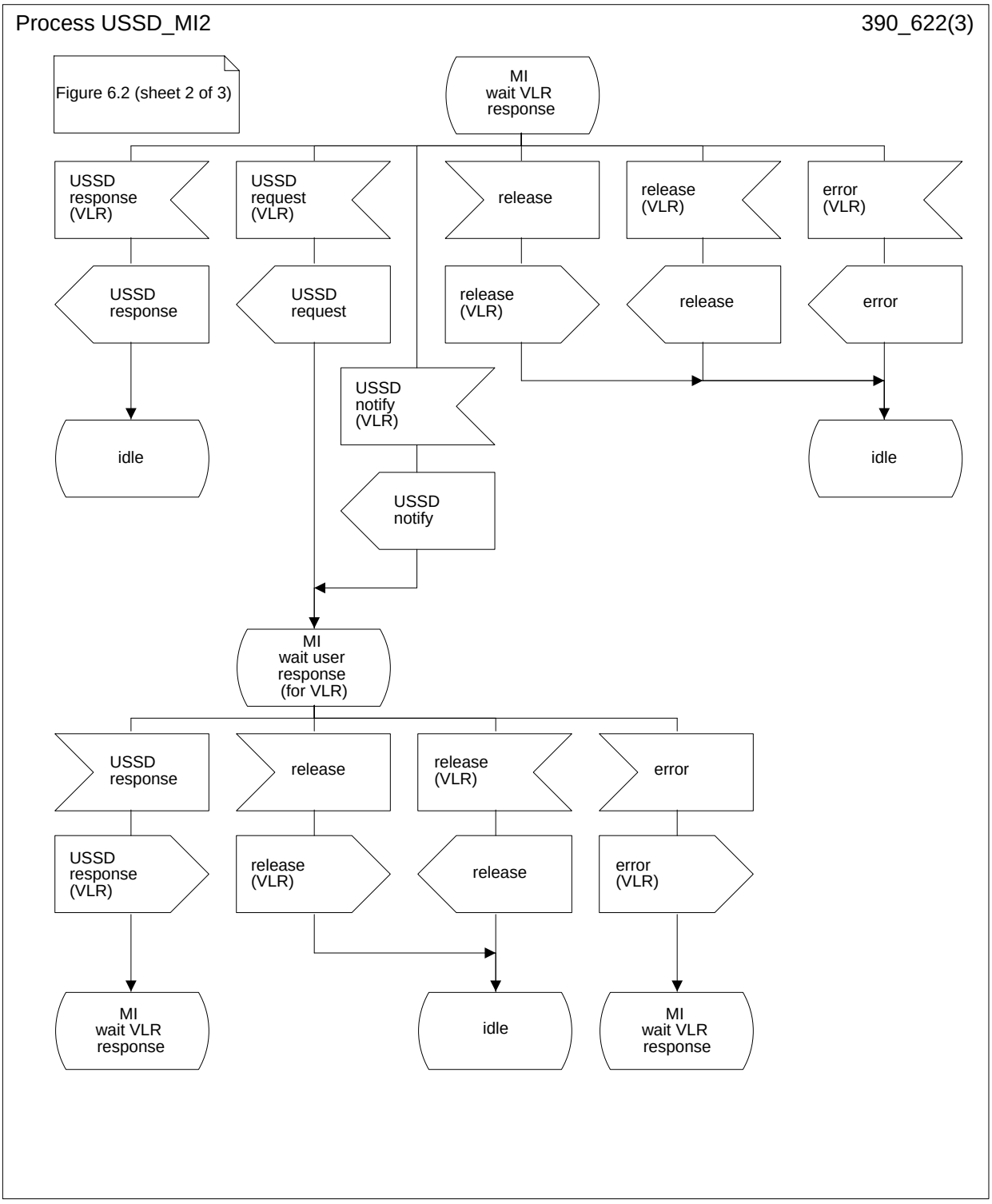



**Figure 6.2 (sheet 2 of 3): Mobile initiated USSD at MSC**

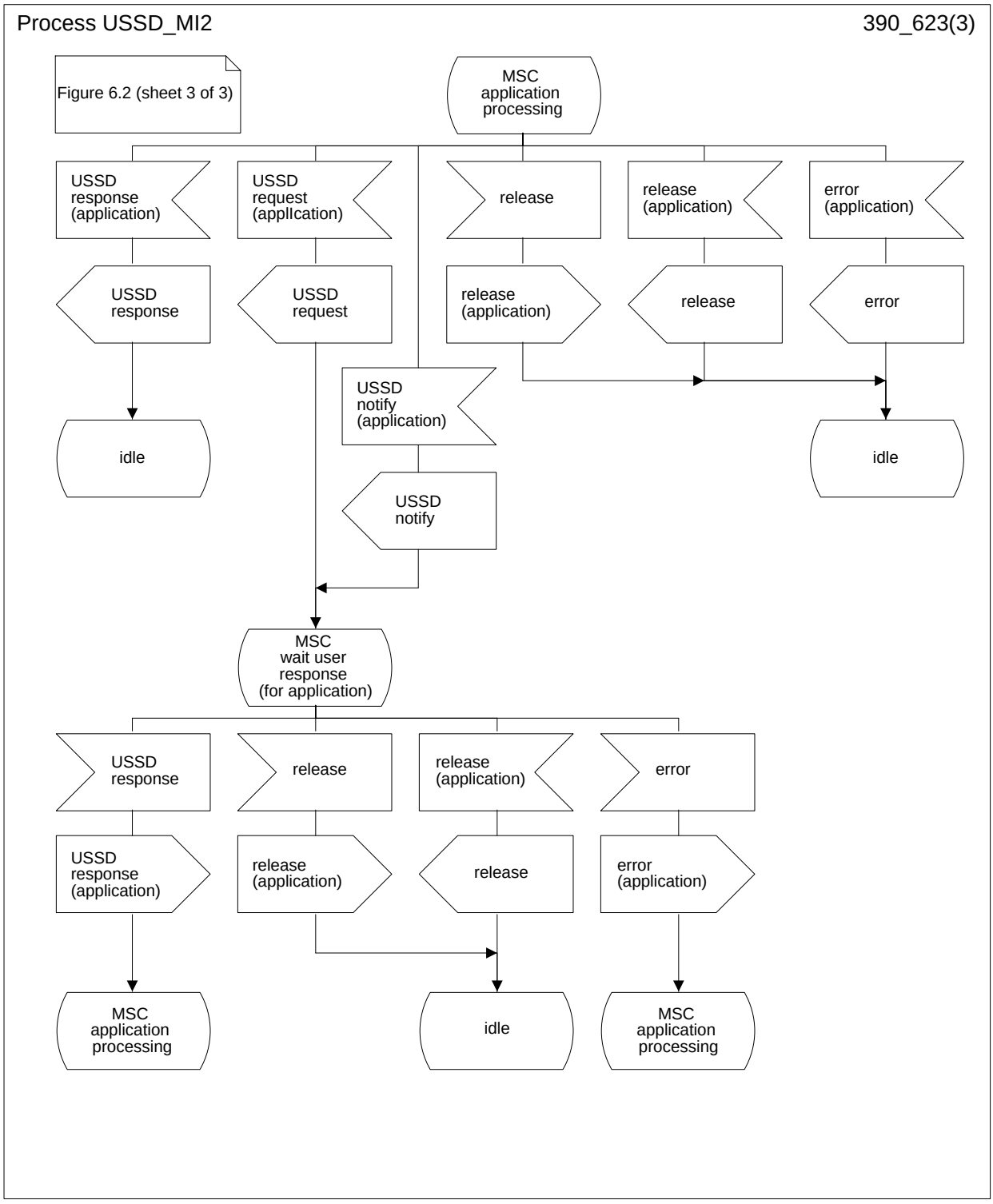



**Figure 6.2 (sheet 3 of 3): Mobile initiated USSD at MSC**

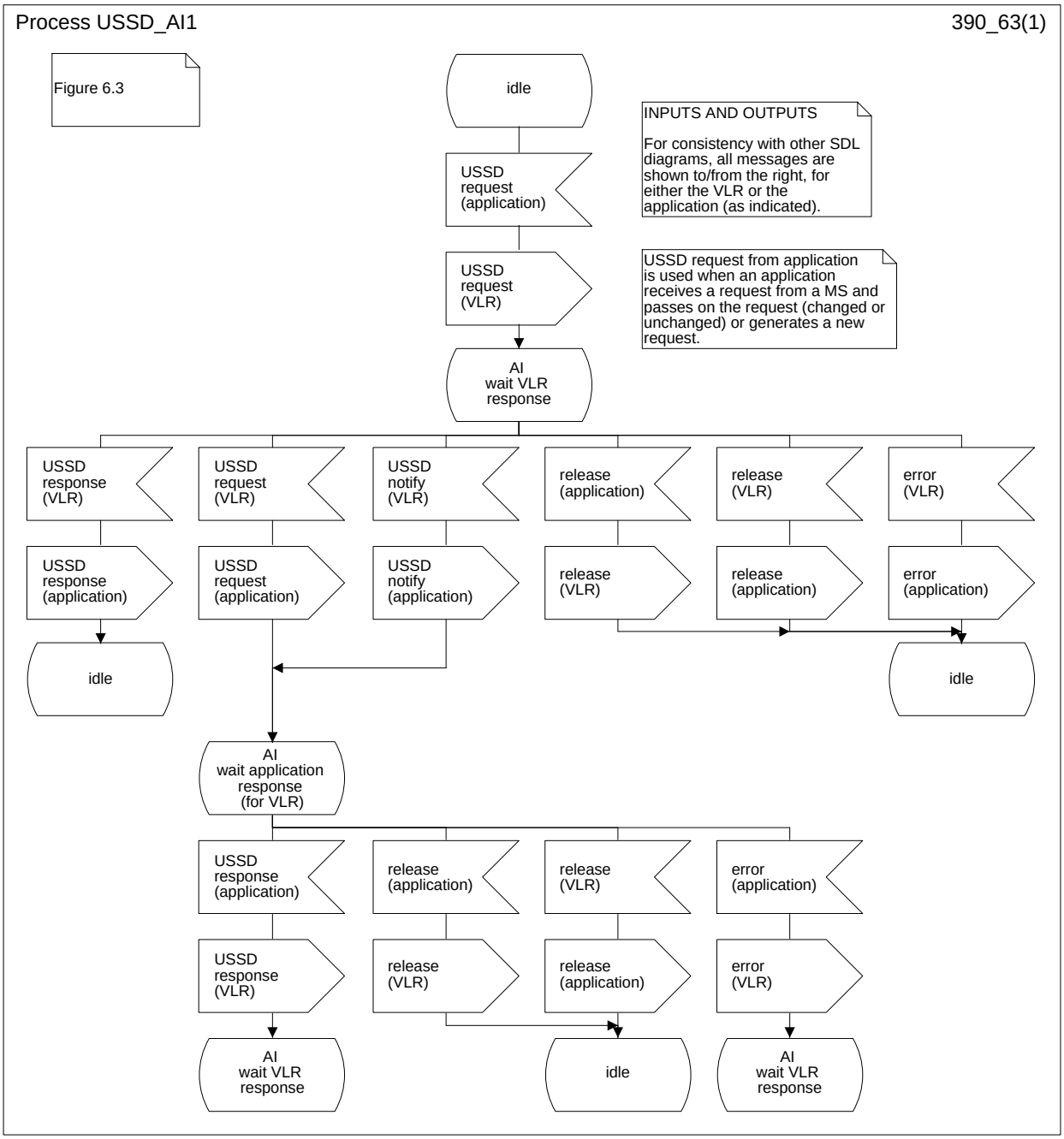



**Figure 6.3: Application initiated USSD at MSC**

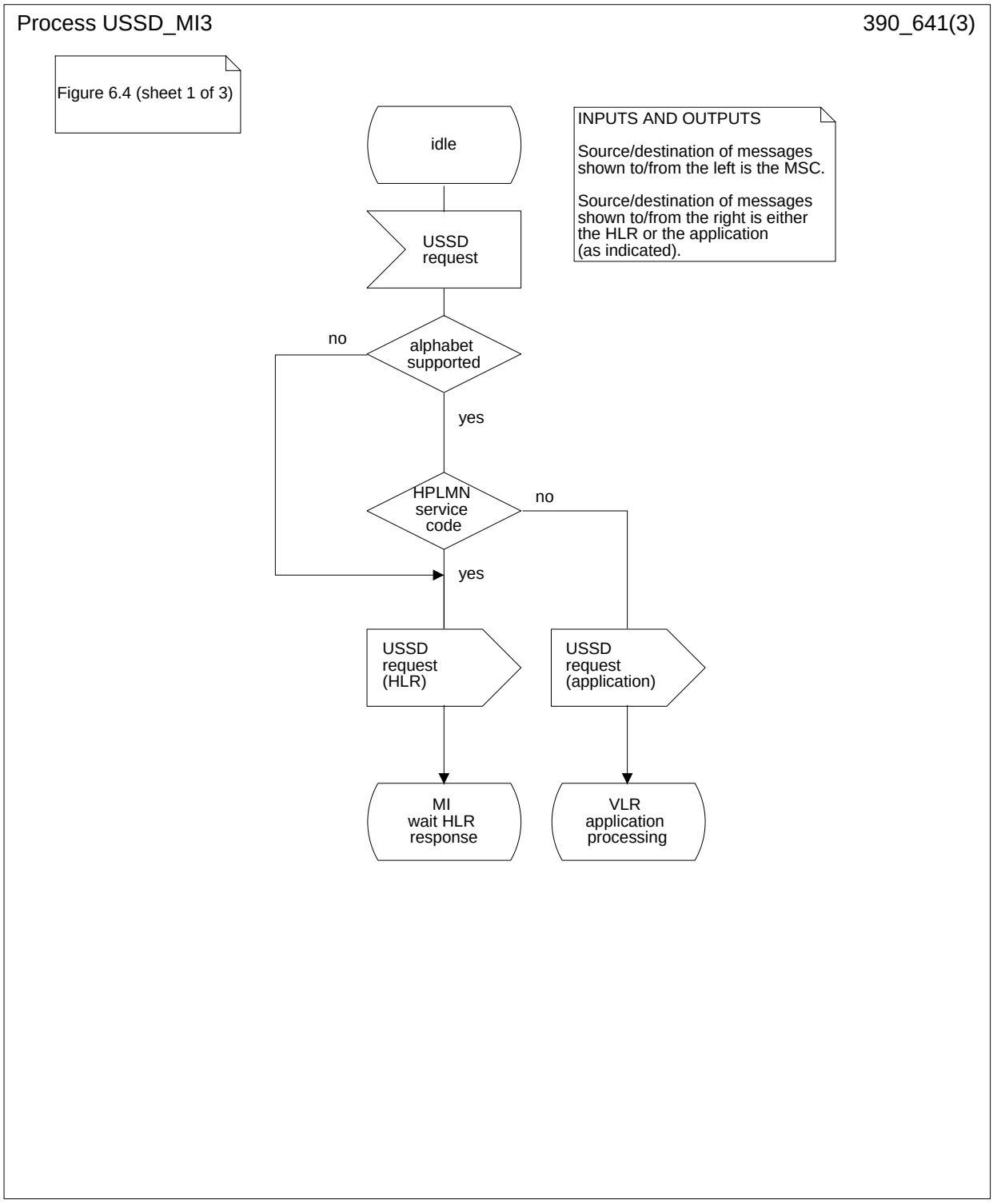



**Figure 6.4 (sheet 1 of 3): Mobile initiated USSD at VLR**

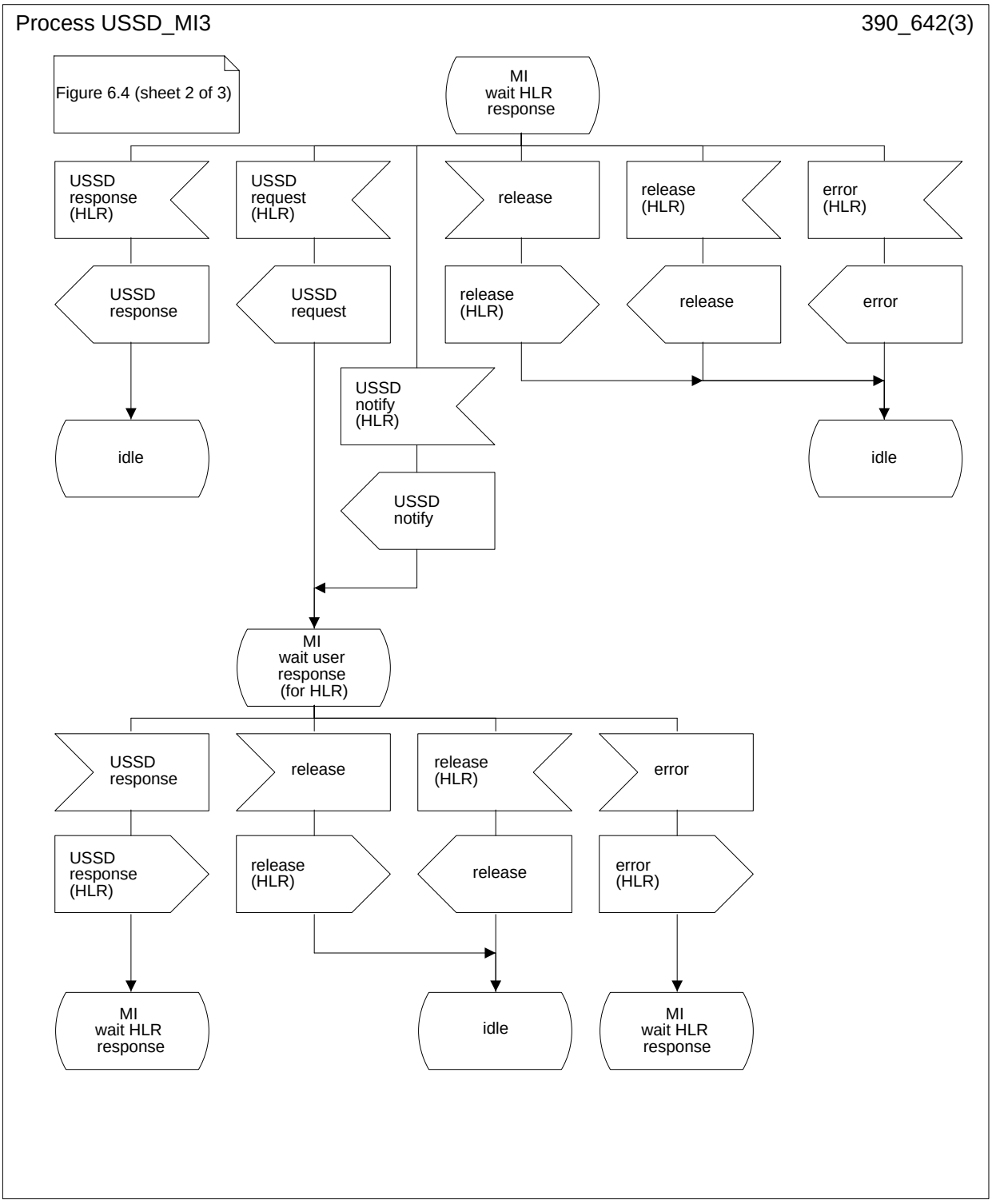



**Figure 6.4 (sheet 2 of 3): Mobile initiated USSD at VLR**

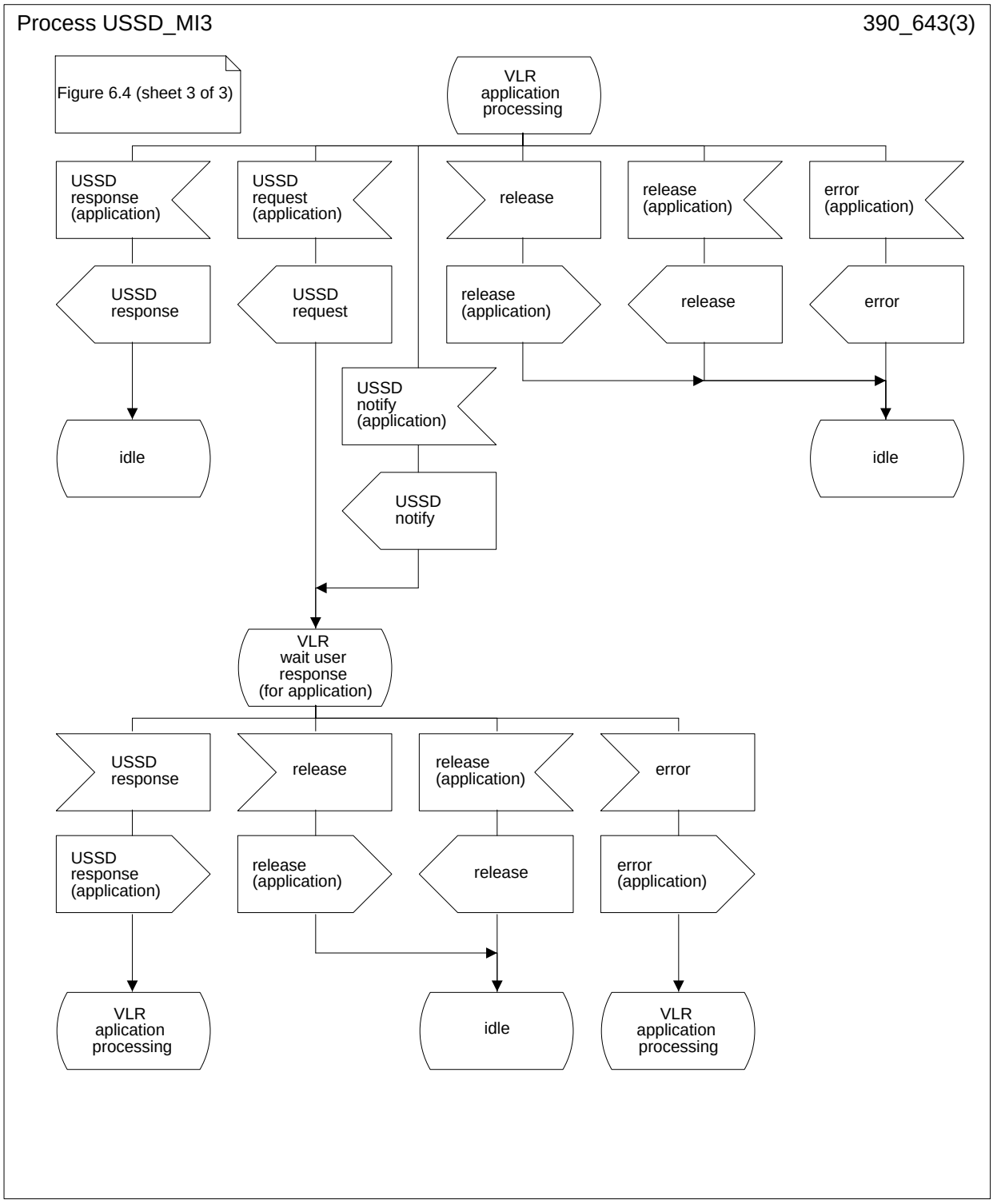



**Figure 6.4 (sheet 3 of 3): Mobile initiated USSD at VLR**

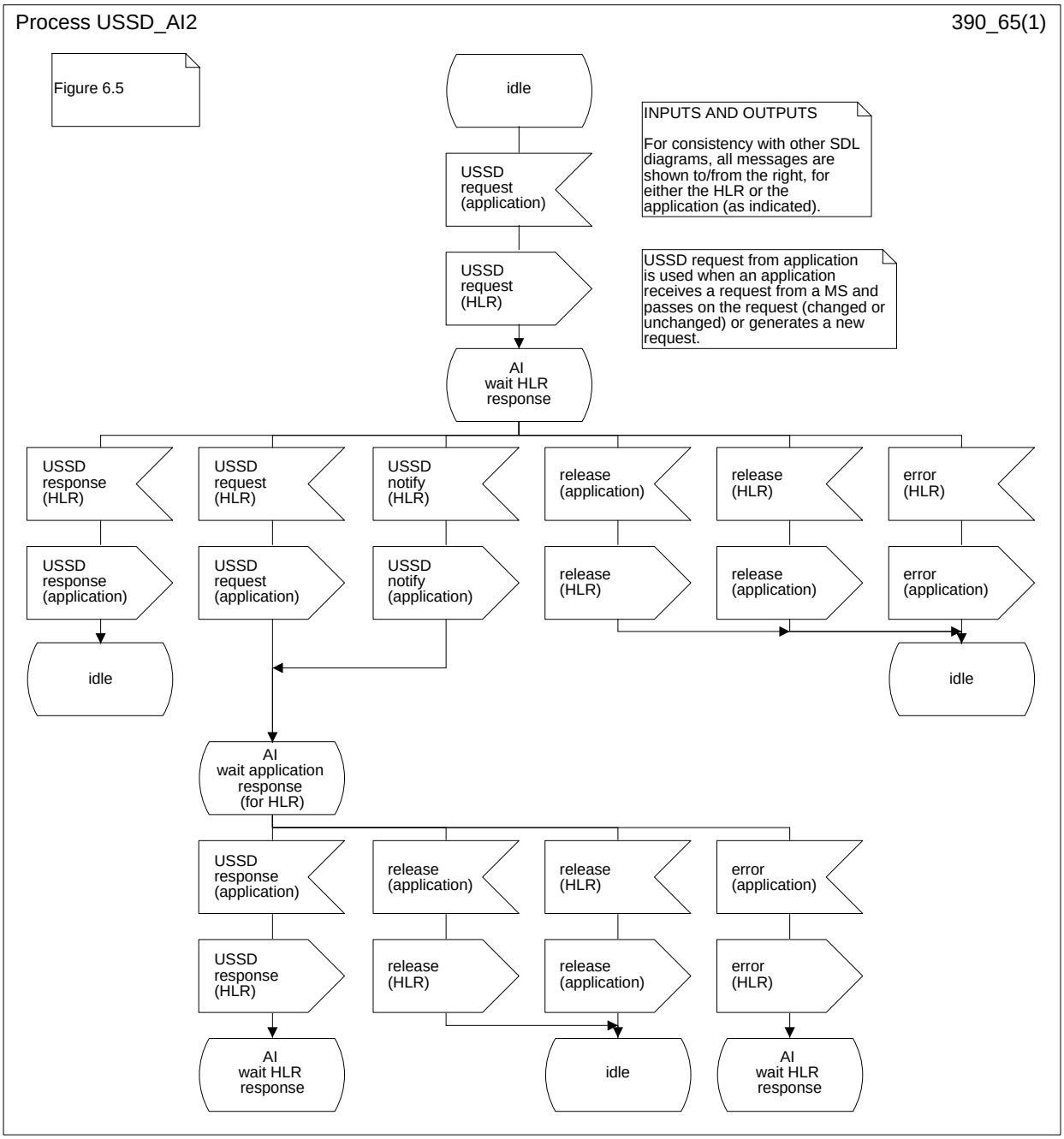



**Figure 6.5: Application initiated USSD at VLR**

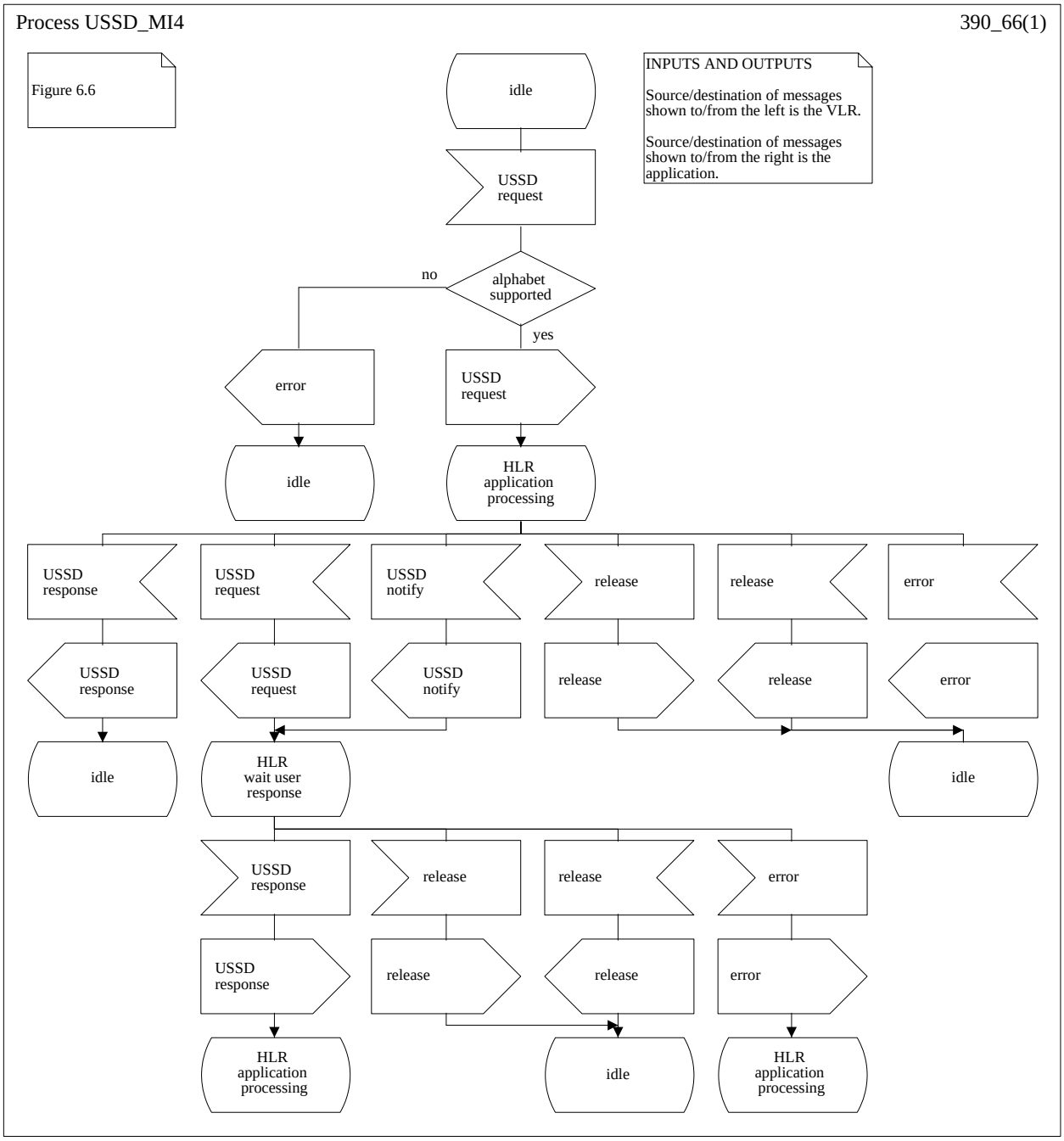
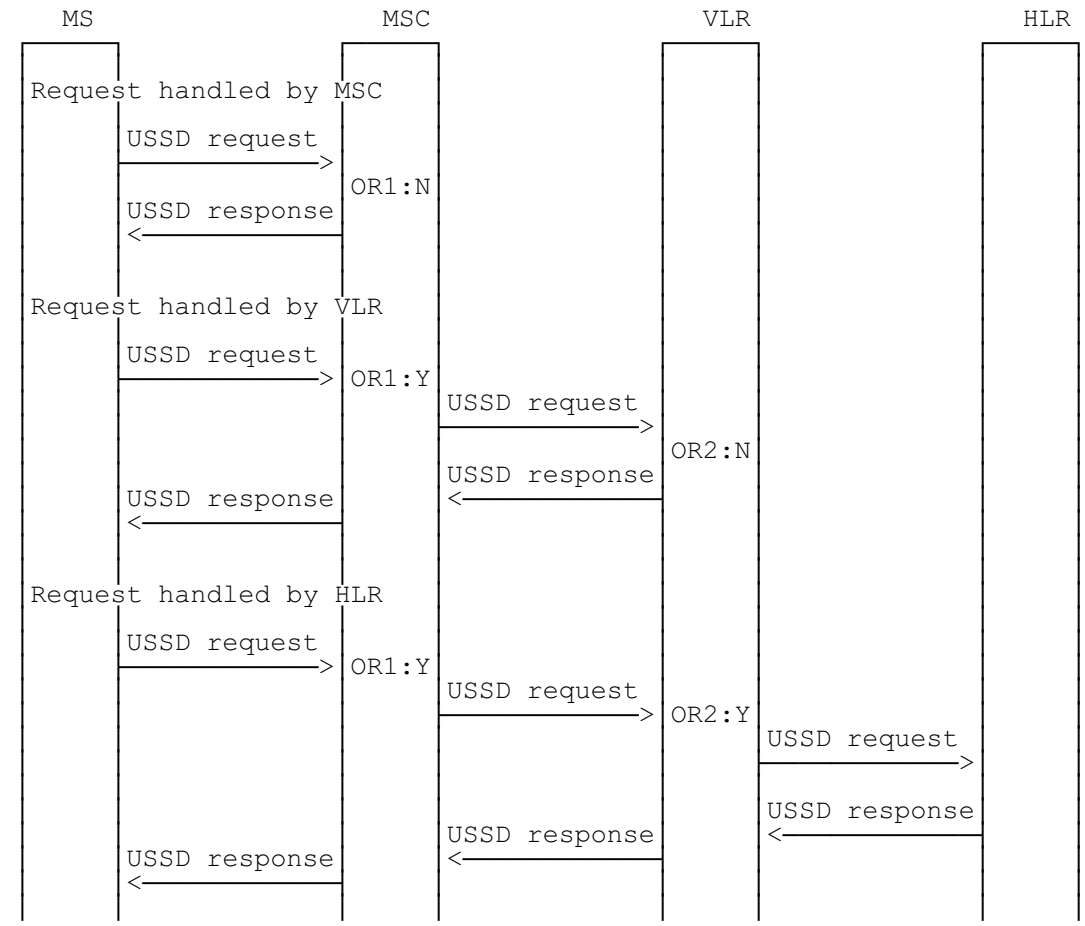


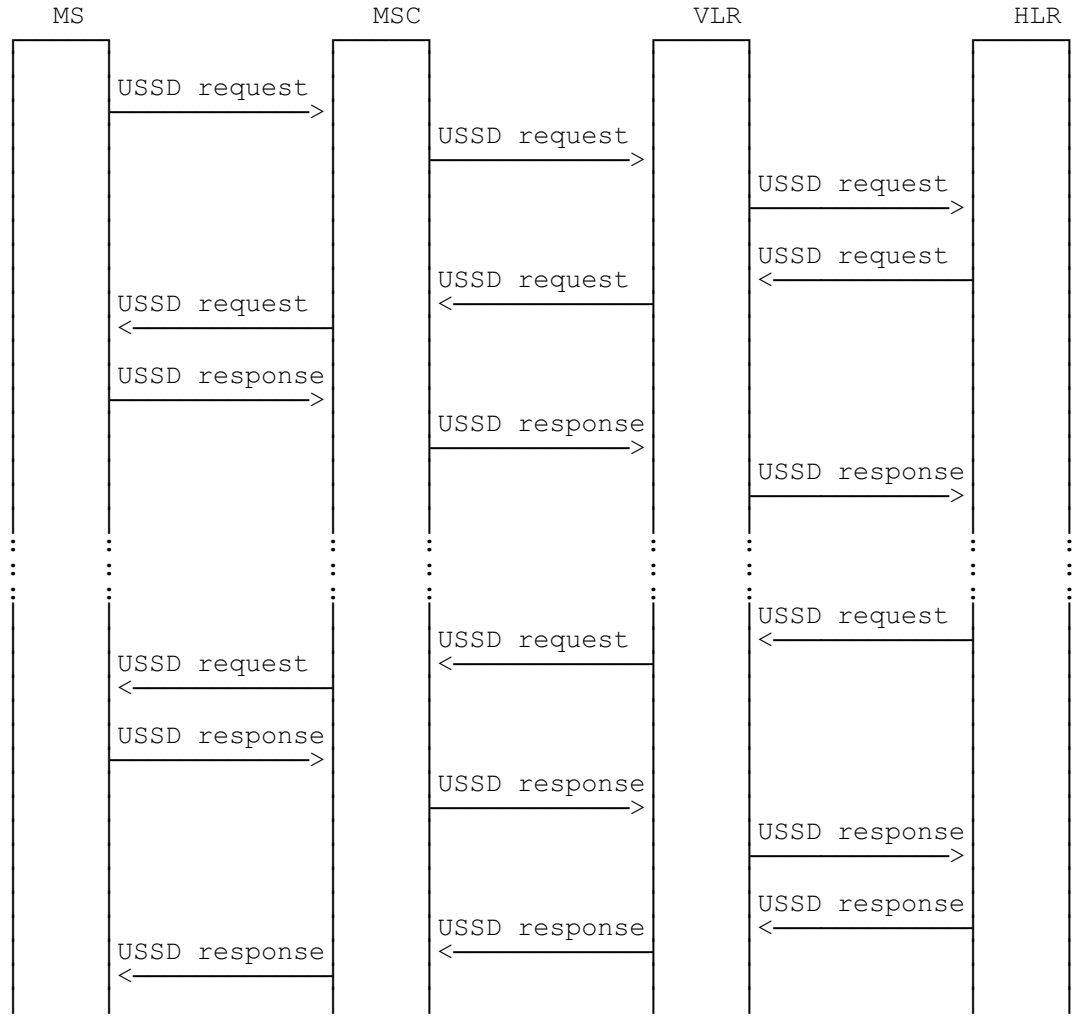

**Figure 6.6: Mobile initiated USSD at HLR**




NOTE: OR1: HPLMN service code Y: Yes
OR2: HPLMN service code and user not in HPLMN N: No

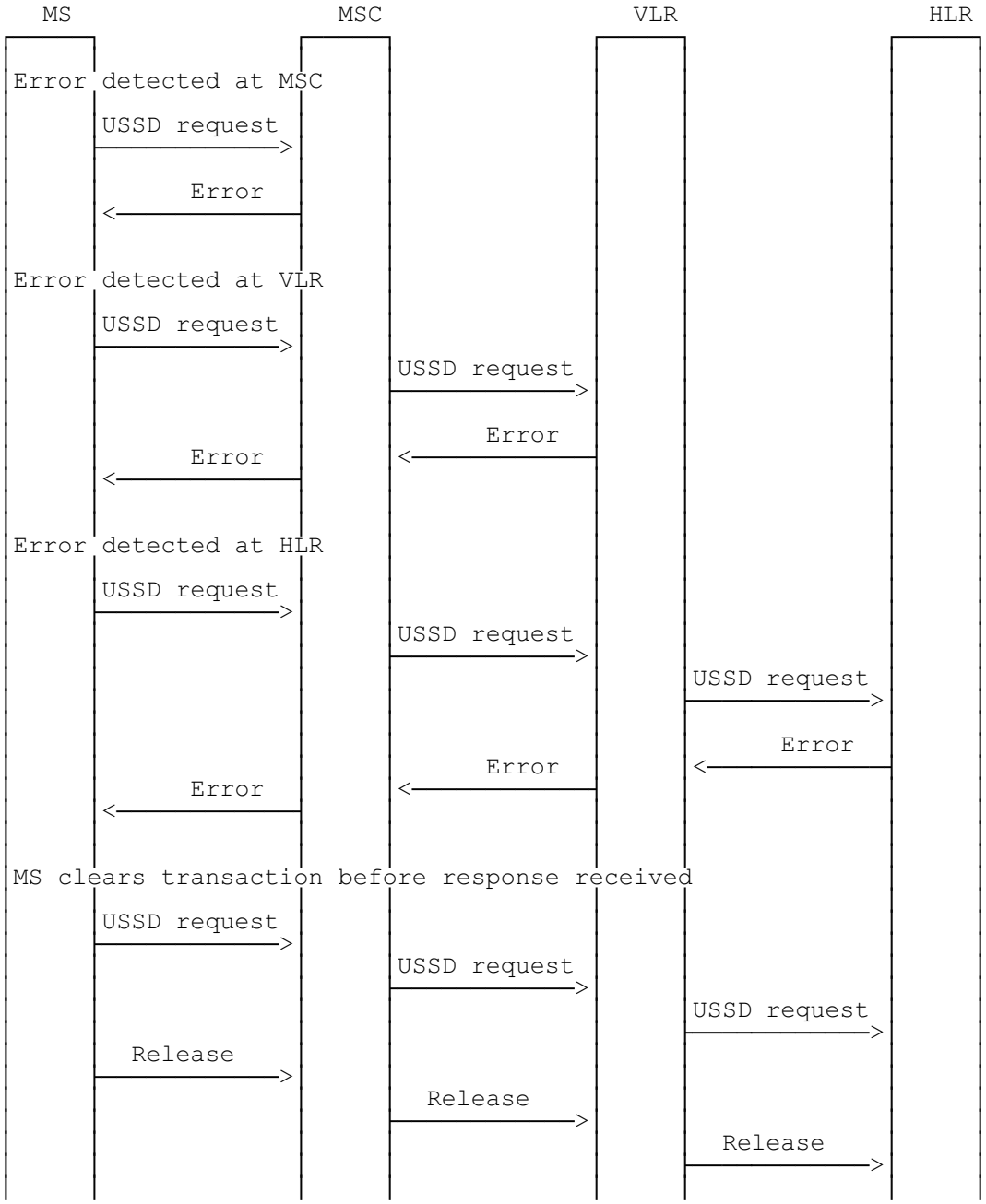
Note that the application at the MSC/VLR may pass the request on to another network entity. This is not shown here.

**Figure 6.7: Information flow for mobile initiated USSD Request (No further information requested)**




NOTE: Note that this call flow only shows one example to illustrate the possible scenarios. See the SDL diagrams for a complete description.

**Figure 6.8: Information flow for mobile initiated USSD Request Handled by HLR, further information requested**




NOTE: This call flow only shows a limited number of examples to illustrate the possible scenarios. See the SDL diagrams for a complete description.

**Figure 6.9: Information flow for mobile initiated failed USSD Request**

### 6.3 Information stored in the HLR

The HLR shall not store any information specific to the use of USSD, although information may be stored for services which are offered by USSD applications.

### 6.4 Information stored in the VLR

The VLR shall not store any information specific to the use of USSD, although information may be stored for services which are offered by USSD applications.

### 6.5 Handover

Handover will have no impact on the operation of this service.

### 6.6 Cross-phase compatibility

If, when a Phase 2 MS sends a mobile initiated USSD request, any network entity is of Phase 1, the request will be rejected. If it is possible to encode the content of the USSD request using the Phase 1 protocol, the MS shall repeat the request, using the Phase 1 protocol.

A Mobile initiated USSD request from a Phase 1 MS uses the Phase 1 protocol. On receipt of such a request, the application shall also use the Phase 1 protocol when sending the response.

A Phase 2 network shall not send network initiated requests or notifications during a mobile initiated USSD request if the MS or any network entity involved in the operation is of Phase 1.

## History

| Document history | |
|---|---|
| December 1996 | Publication of GSM 03.90 version 5.0.0 |
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